BIẾN ĐỔI LAPLACE NGƯỢC

PHÂN TÍCH PHÂN THỨC PHỨC TẠP THÀNH PHÂN THỨC ĐƠN GIẢN

Giả sử: $\boldsymbol{F(s) = \frac{N(s)}{D(s)}}$

N(s), D(s) là các đa thức

Tất cả các nghiệm của phương trình N(s) = 0 được gọi là điểm zeros

Tất cả các nghiệm của phương trình D(s) = 0 được gọi là điểm cực

Nếu bậc N(s) < bậc của D(s).

Trường hợp 1: Mẫu số chỉ có nghiệm đơn

$$\boldsymbol{F(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{(s+p_1)(s+p_2)\ldots(s+p_n)}}$$

Ta phân tích thành: $F(s) = \frac{N(s)}{D(s)} = \frac{k_1}{(s+p_1)} + \frac{k_2}{(s+p_2)} + \ldots + \frac{k_n}{(s+p_n)}$

Với $k_i = (s + p_i).F(s)\Big|_{s=-p_i}$

$$L^{-1}\left[\frac{k}{(s+a)}\right] = k.e^{-at}$$

$$\rightarrow \boldsymbol{f(t) = (k_1.e^{-p_1 t} + k_2.e^{-p_2 t} + \cdots + (k_n.e^{-p_n t})}$$

Ví dụ: $$F(s) = \frac{s^2 + 12}{s(s+2)(s+3)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+3}$$

METHOD 1 Residue method:

$$A = sF(s)\,|_{s=0} = \left.\frac{s^2 + 12}{(s+2)(s+3)}\right|_{s=0} = \frac{12}{(2)(3)} = 2$$

$$B = (s+2)F(s)\,|_{s=-2} = \left.\frac{s^2 + 12}{s(s+3)}\right|_{s=-2} = \frac{4 + 12}{(-2)(1)} = -8$$

$$C = (s+3)F(s)\,|_{s=-3} = \left.\frac{s^2 + 12}{s(s+2)}\right|_{s=-3} = \frac{9 + 12}{(-3)(-1)} = 7$$

$$\Rightarrow \quad F(s) = \frac{2}{s} - \frac{8}{s+2} + \frac{7}{s+3}$$

Ví dụ: $F(s) = \dfrac{s^2 + 12}{s(s+2)(s+3)} = \dfrac{A}{s} + \dfrac{B}{s+2} + \dfrac{C}{s+3}$

METHOD 2 Algebraic method:

Cân bằng 2 vế: $s^2 + 12 = A(s+2)(s+3) + Bs(s+3) + Cs(s+2)$

Constant: $12 = 6A \quad \Rightarrow \quad A = 2$

s: $0 = 5A + 3B + 2C \quad \Rightarrow \quad 3B + 2C = -10$

s^2: $1 = A + B + C \quad \Rightarrow \quad B + C = -1$

$$\Rightarrow \quad F(s) = \frac{2}{s} - \frac{8}{s+2} + \frac{7}{s+3}$$

$$y'' + ay' + by = r(t),$$

$$[s^2Y - sy(0) - y'(0)] + a[sY - y(0)] + bY = R(s)$$

$$(s^2 + as + b)Y = (s + a)y(0) + y'(0) + R(s).$$

Ví dụ: $y'' - y' - 6y = 0$; $y(0) = 11$; $y'(0) = 28$

$(s^2 - s - 6)\,Y = (s - 1)11 + 28$

$Y = (11s + 17)/(s-3)(s+2) = 10/(s-3) + 1/(s+2)$

$y = L^{-1}(Y) = 10e^{3t} + e^{-2t}$

Ví dụ: $y'' + 7y' + 12y = 21e^{3t}$; $y(0) = 3{,}5$; $y'(0) = -10$

$(s^2 + 7s + 12)\,Y = (s + 7)3{,}5 - 10 + 21/(s-3)$

$Y = (1/2)x1/(s-3) + (1/2)x1/(s+3) + (5/2)x1/(s+4)$

$y = L^{-1}(Y) = (1/2)e^{3t} + (1/2)e^{-3t} + (5/2)e^{-4t}$

Trường hợp 2: mẫu số có nghiệm bội bậc n

$$\boldsymbol{F(s)} = \frac{\boldsymbol{N(s)}}{\boldsymbol{D(s)}} = \frac{k_n}{(s+p)^n} + \frac{k_{n-1}}{(s+p)^{n-1}} + \ldots + \frac{k_2}{(s+p)^2} + \frac{k_1}{(s+p)} + F_1(s)$$

Với: $k_n = \left[(s+p)^n . F(s)\right]\Big|_{s=-p}$

$k_{n-m} = \frac{1}{m!}\frac{d^m}{ds^m}\left[(s+p)^n . F(s)\right]\Big|_{s=-p}$; với m=1,2,3…,n-1

$$L^{-1}\left[\frac{1}{(s+a)^n}\right] = \frac{t^{n-1}.e^{-at}}{(n-1)!}$$

$$\rightarrow \boldsymbol{f(t)} = \left(\boldsymbol{k_1.e^{-pt} + k_2.t.e^{-pt} + \frac{k_3.t^2.e^{-pt}}{2!} + \cdots + \frac{k_n.t^{n-1}.e^{-pt}}{(n-1)!}}\right) + \boldsymbol{f_1(t)}$$

Ví dụ: $$V(s) = \frac{10s^2 + 4}{s(s+1)(s+2)^2}$$

$$= \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+2)^2} + \frac{D}{s+2}$$

METHOD 1 Residue method:

$$A = sV(s)\,|_{s=0} = \frac{10s^2 + 4}{(s+1)(s+2)^2}\bigg|_{s=0} = \frac{4}{(1)(2)^2} = 1$$

$$B = (s+1)V(s)\,|_{s=-1} = \frac{10s^2 + 4}{s(s+2)^2}\bigg|_{s=-1} = \frac{14}{(-1)(1)^2} = -14$$

$$C = (s+2)^2V(s)\,|_{s=-2} = \frac{10s^2 + 4}{s(s+1)}\bigg|_{s=-2} = \frac{44}{(-2)(-1)} = 22$$

$$D = \frac{d}{ds}[(s+2)^2V(s)]\bigg|_{s=-2} = \frac{d}{ds}\left(\frac{10s^2 + 4}{s^2 + s}\right)\bigg|_{s=-2}$$

$$= \frac{(s^2 + s)(20s) - (10s^2 + 4)(2s + 1)}{(s^2 + s)^2}\bigg|_{s=-2} = \frac{52}{4} = 13$$

METHOD 2 Algebraic method:

$$10s^2 + 4 = A(s+1)(s+2)^2 + Bs(s+2)^2 + Cs(s+1) + Ds(s+1)(s+2)$$

Constant: $4 = 4A \quad \Rightarrow \quad A = 1$

s: $0 = 8A + 4B + C + 2D \quad \Rightarrow \quad 4B + C + 2D = -8$

s^2: $10 = 5A + 4B + C + 3D \quad \Rightarrow \quad 4B + C + 3D = 5$

s^3: $0 = A + B + D \quad \Rightarrow \quad B + D = -1$

Solving these simultaneous equations gives $A = 1, B = -14, C = 22, D = 13$, so that

$$V(s) = \frac{1}{s} - \frac{14}{s+1} + \frac{13}{s+2} + \frac{22}{(s+2)^2}$$

$$y'' + ay' + by = r(t).$$

$$[s^2Y - sy(0) - y'(0)] + a[sY - y(0)] + bY = R(s)$$

$$(s^2 + as + b)Y = (s + a)y(0) + y'(0) + R(s).$$

Ví dụ: $y'' - 4y' + 4y = 0, \quad y(0) = 8{,}1\,, y'(0) = 3{,}9$

Apply Laplace transform with $\mathcal{L}\{y\} = Y$, we get:

$$(s^2 - 4s + 4)Y = (s - 4).8{,}1 + 3{,}9$$

$$\Rightarrow Y = \frac{8{,}1s - 28{,}5}{(s-2)^2} = \frac{8{,}1}{s-2} - \frac{12{,}3}{(s-2)^2}$$

$$y = \mathcal{L}^{-1}\{Y\} = 8{,}1e^{2t} - 12{,}3te^{2t}$$

Trường hợp 3: mẫu số có nghiệm phức

$$\boldsymbol{F(s)} = \frac{\boldsymbol{N(s)}}{\boldsymbol{D(s)}} = \frac{A_1 s + A_2}{s^2 + as + b} + F_1(s) = \frac{A_1(s+\alpha) + B_1\beta}{(s+\alpha)^2 + \beta^2} + F_1(s)$$

$$= \frac{A_1(s+\alpha)}{(s+\alpha)^2 + \beta^2} + \frac{B_1\beta}{(s+\alpha)^2 + \beta^2} + F_1(s)$$

$$\rightarrow \boldsymbol{f(t) = (\,A_1 . e^{-\alpha t} . cos(\beta t) + B_1 . e^{-\alpha t} . sin(\beta t)) + f_1(t)}$$

Ví dụ: $H(s) = \dfrac{20}{(s+3)(s^2+8s+25)} = \dfrac{A}{s+3} + \dfrac{Bs+C}{(s^2+8s+25)}$

METHOD 1 **Combination of methods:**

$$A = (s+3)H(s)\,|_{s=-3} = \left.\frac{20}{s^2+8s+25}\right|_{s=-3} = \frac{20}{10} = 2$$

Tìm B và C bằng cách cân bằng 2 vế hoặc cho các giá trị của s (s không phải điểm cực của H(s)) để tìm B và C.

Tìm C: cho s=0, ta được: $\dfrac{20}{75} = \dfrac{A}{3} + \dfrac{C}{25}$ $\Rightarrow$ $C = -10$

Tìm B: cho s=1, ta được: $\dfrac{20}{(4)(34)} = \dfrac{A}{4} + \dfrac{B+C}{34}$ $\Rightarrow$ $B = -2$

Ví dụ: $H(s) = \dfrac{20}{(s+3)(s^2+8s+25)} = \dfrac{A}{s+3} + \dfrac{Bs+C}{(s^2+8s+25)}$

METHOD 2 Algebraic method:

$$\begin{aligned} 20 &= A(s^2+8s+25) + (Bs+C)(s+3) \\ &= A(s^2+8s+25) + B(s^2+3s) + C(s+3) \end{aligned} \qquad \textbf{(15.11.4)}$$

Equating coefficients gives

s^2: $\quad 0 = A + B \quad \Rightarrow \quad A = -B$

s: $\quad 0 = 8A + 3B + C = 5A + C \quad \Rightarrow \quad C = -5A$

Constant: $\quad 20 = 25A + 3C = 25A - 15A \quad \Rightarrow \quad A = 2$

That is, $B = -2$, $C = -10$. Thus,

$$\begin{aligned} H(s) &= \frac{2}{s+3} - \frac{2s+10}{(s^2+8s+25)} = \frac{2}{s+3} - \frac{2(s+4)+2}{(s+4)^2+9} \\ &= \frac{2}{s+3} - \frac{2(s+4)}{(s+4)^2+9} - \frac{2}{3}\frac{3}{(s+4)^2+9} \end{aligned}$$

$$y'' + ay' + by = r(t),$$

$$[s^2Y - sy(0) - y'(0)] + a[sY - y(0)] + bY = R(s)$$

$$(s^2 + as + b)Y = (s + a)y(0) + y'(0) + R(s).$$

Ví dụ: $y'' + 9y = 10e^{-t}, \qquad y(0) = y'(0) = 0$

Apply Laplace transform with $\mathcal{L}\{y\} = Y$, we get:

$$(s^2 + 9)Y = \frac{10}{s+1} \Rightarrow Y = \frac{10}{(s+1)(s^2+9)} = \frac{1}{s+1} - \frac{s}{s^2+9} + \frac{1}{s^2+9}$$

$$\Rightarrow y = \mathcal{L}^{-1}\{Y\} = e^{-t} - \cos 3t + \frac{1}{3}\sin 3t$$

Ví dụ: $y'' - 6y' + 5y = 29cos\, 2t, \qquad y(0) = 3{,}2,\ y'(0) = 6{,}2$

Apply Laplace transform with $\mathcal{L}\{y\} = Y$, we get:

$$(s^2 - 6s + 5)Y = (s-6).3{,}2 + 6{,}2 + \frac{29s}{s^2+4}$$

$$\Rightarrow Y = \frac{(3{,}2s - 13)(s^2+4) + 29s}{(s^2+4)(s-5)(s-1)}$$

$$= \frac{1}{5}.\frac{s}{s^2+4} - \frac{24}{5}.\frac{1}{s^2+4} + \frac{2}{s-5} + \frac{1}{s-1}$$

$$\Rightarrow y = \mathcal{L}^{-1}\{Y\} = \frac{1}{5}\cos 2t - \frac{12}{5}\sin 2t + 2e^{5t} + e^{t}$$